香港資優教育學苑
The Hong Kong Academy for Gifted Education

IMO Training 2020 –21 (Phase I) Level 1

Algebra III (代數 III)

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Basic Inequalities (基本不等式)

There are two types of inequality problems:

<u>Type 1</u>: Conditional Inequalities (條件不等式):

A Conditional Inequality is an inequality which is valid only when the variables(s) in the inequality takes certain values.

(For example: $3x + 4 > 19 - 2x$ is valid only when $x > 3$.)

<u>Ex. 1</u>: Solve the inequality $2x^2 - 7x + 2 \geq 0$

<u>Ex. 2</u>: Solve the inequality $\dfrac{x^2 - 9x + 11}{x^2 - 2x + 1} \geq 7$

<u>Ex. 3</u>: Solve the inequality $\sqrt{4 - x^2} > x + \sqrt{3} - 1$.

<u>Ex. 4</u>: Solve the inequalities $\frac{1}{x+1} + \frac{1}{x+4} > \frac{1}{x+2} + \frac{1}{x+3}$

<u>Ex. 5</u>: (Second IMO (1960) No. 2.)

For what values of the variable x does the following inequality hold?

$$\frac{4x^2}{(1-\sqrt{1+2x})^2} < 2x + 9$$

<u>Ex. 6</u>: (Fourth IMO (1962) No. 2.)

Determine all real numbers x which satisfy the inequality:

$$\sqrt{3-x} - \sqrt{x+1} > \frac{1}{2}$$

<u>Type 2</u>: Absolute Inequalities (絕對不等式):

An absolute inequality is an inequality which is valid no matter what values the variable(s) assume in the permitted number set.

(For example: For any positive real numbers a and b, $a + b \geq 2\sqrt{ab}$.)

(<u>Proof</u>: $(\sqrt{a} - \sqrt{b})^2 \geq 0$, hence $a + b \geq 2\sqrt{ab}$.)

<u>Ex. 7</u>: For any real numbers a and b such that a>b, prove that $a^3 > b^3$.

<u>Ex. 8</u>: For any positive real numbers a , b and c, $a^3 + b^3 + c^3 \geq 3abc$

<u>Ex. 9</u>: (Sixth IMO (1964) No. 2.) Suppose a, b, a are the sides of a triangle, prove that

$$a^2(b+c-a) + b^2(c+a-b) + c^2(a+b-c) \leq 3abc$$

Fundamental Rules

For any real numbers a, b, c and n:

1. **(i) $a > b$ if and only if $a - b$ is positive,**

Ex. 10: If x is real, which is larger, $x^2 + 3$ or $3x$?

Ex. 11: Given that a, b, c are real numbers such that $a+b+c = 1$, prove that $a^2+b^2+c^2 \geq \frac{1}{3}$.

2. **(i) If $\frac{a}{b} > 1$ and $b > 0$ then $a > b$,**

Ex. 12: If $a>b>0$, which is larger, $\frac{a^2-b^2}{a^2+b^2}$, or $\frac{a-b}{a+b}$?

(ii) If $\frac{a}{b} = 1$ and $b > 0$ then $a = b$,

(iii) If $\frac{a}{b} < 1$ and $b > 0$ then $a < b$.

3. **If $a > b$ and $b > c$ then $a > c$.** (Transitive property 傳遞性)

4. **If $a > b$ then $a + c > b + c$. (Here c may be positive or negative.)**

5. **(i) If $a > b$ and $c > 0$ then $ac > bc$,**
(ii) If $a > b$ and $c < 0$ then $ac < bc$.

6. **If $a > b$ and $c > d$ then $a + c > b + d$.**
(Note: The converse (逆定理) is not necessarily true.)

7. **If $a > b > 0$ and $c > d > 0$ then $ac > bd$.**
(Note: The converse (逆定理) is not necessarily true.)

8. **(i) If $a > b > 0$ and $m > 0$, then $\frac{a}{b} > \frac{a+m}{b+m}$.**
(ii) If $b > a > 0$ and $m > 0$, then $\frac{a}{b} < \frac{a+m}{b+m}$.

Ex. 13: If $\frac{1}{2} < a < 1$ and $\frac{1}{2} < b < 1$, prove that $\frac{1}{a+1} + \frac{1}{b+1} < \frac{3}{2}$.

Ex. 14 : (1974 IMO No. 5): Determine all possible values of

$$\frac{a}{a+b+d}+\frac{b}{b+c+a}+\frac{c}{c+d+b}+\frac{d}{d+a+c}$$, where a, b, c, d are positive real numbers.

9. **If $a > b > 0$ and $n > 0$ then $a^n > b^n$, and hence $a^{-n} < b^{-n}$.**

Note: Consider the case $n=\frac{1}{2}$, in this case the theorem means $a > b > 0 \Rightarrow \sqrt{a} > \sqrt{b}$.

In other words, under the condition that $a>0$ and $b>0$, $a>b \Leftrightarrow a^2 > b^2$.

Ex. 15: Prove that $2\sqrt{2} - \sqrt{5} > 2\sqrt{3} - 3$

Ex. 16: Given $m > 1$. Find out which is larger, $\sqrt{m+1} - \sqrt{m}$, or $\sqrt{m} - \sqrt{m-1}$.

10. **(i) If $ab > 0$ then either ($a > 0$ and $b > 0$) or ($a < 0$ and $b < 0$),**
(ii) If $ab < 0$ then either ($a > 0$ and $b < 0$) or ($a < 0$ and $b > 0$).

11. **For any real numbers a, $a^2 \geqq 0$.**
(Note that $a^2 \geqq 0$ does not imply that $a \geqq 0$.)

Ex. 17 : Prove that for any positive real number x, $x + \frac{1}{x} \geqq 2$

12. **The quadratic equation $ax^2 + bx + c = 0$, where $a \neq 0$, gives real roots in x if and only if $b^2 - 4ac \geqq 0$.**

13. **The function $f(x) = ax^2 + bx + c$, where $a \neq 0$, assumes positive values for all values of x if and only if $a > 0$ and $b^2 - 4ac < 0$.**

Ex. 18: If $a>0$, prove that $2a^3 + a^2 + 4 > 4a$.

14. **$|a| \leqq |b|$ if and only if $-|b| \leqq a \leqq |b|$.**

15. **$||a| - |b|| \leqq |a + b| \leqq |a| + |b|$.**

16. **$|a_1 + a_2 + a_3 + \ldots\ldots + a_n| \leqq |a_1| + |a_2| + |a_3| + \ldots\ldots + |a_n|$.**

Methods used in proving absolute inequalities:

1. Direct Comparison (比較法)
   (Using Fundamental Rule 1 and Rule 2)
   Ex. 19: If x is real prove that $(x-3)(x-1)(x+2)(x+4) + 26$ is always positive.

   Ex. 20: Prove that if a, b, c > 0, $ca^2 + \frac{b}{c} \geq 2a\sqrt{b}$

   Ex. 21: If a is a positive real numbers, prove that $a^3 + 2 \geq a^2 + 2\sqrt{a}$.

   Ex. 22: If no two of the three positive real numbers a, b and c are equal, prove that
   (i) $a^a b^b > a^b b^a$.
   (ii) $a^a b^b c^c > a^c b^a c^b$

   Ex. 23: If a, b, c are distinct positive real numbers, prove that $a^a b^b c^c > abc^{\frac{a+b+c}{3}}$.

2. Forward deduction (綜合法)
   (Using Fundamental Rule 6 and Rule 7)
   Ex. 24: If a, b, c are distinct positive real numbers, prove that
   $$\frac{b+c-a}{a} + \frac{c+a-b}{b} + \frac{a+b-c}{c} > 3$$
   Ex. 25: Given a, b and c are real numbers. Prove that
   $$\sqrt{a^2+b^2} + \sqrt{b^2+c^2} + \sqrt{c^2+a^2} \geq \sqrt{2}\,(a+b+c)$$
   Ex. 26: If a, b, c and x, y, z are real numbers such that $a^2 + b^2 + c^2 = 1$ and $x^2 + y^2 + z^2 = 1$, prove that $|ax + by + cz| \leq 1$.

3. Backward Trace (反推法)
   (Working backwards, should make sure that the use of the symbol ⇔ is valid.)
   Ex. 27: Given that a, b are real numbers and a>b, prove that $\frac{a}{2} \geq \sqrt{2a(a-b)} - (a-b)$
   Ex. 28: Given that a>3, prove that $\sqrt{a-3} - \sqrt{a-1} < \sqrt{a-2} - \sqrt{a}$.
   Ex. 29: If a, b, c are positive real numbers, prove that
   $$\frac{a+b}{c} + \frac{b+c}{a} + \frac{c+a}{b} \geq 6.$$
   Ex. 30: Let $a_1, a_2, \ldots, a_n$ be real numbers greater than 1. Furthermore, for any k, $1 \leq k \leq n-1$, $|a_{k+1} - a_k| < 1$.
   Prove that $\frac{a_1}{a_2} + \frac{a_2}{a_3} + \ldots + \frac{a_{n-1}}{a_n} + \frac{a_n}{a_1} < 2n - 1$

4. Expanding and Grouping Fractions (放縮法)
   (Used when dealing with sum of series.)
   Ex. 31: For any positive integer n, prove that
   $$1 + \frac{1}{2^2} + \frac{1}{3^2} + \ldots + \frac{1}{n^2} > \frac{3n+1}{2n+2}$$
   Ex. 32: Prove that for any positive integer n,
   $$1 < \frac{1}{n+1} + \frac{1}{n+2} + \ldots + \frac{1}{3n+1} < 2$$
   Ex. 33: Prove that $16 < \sum_{k=1}^{80} \frac{1}{\sqrt{k}} < 17$.
   Ex. 34: For any positive integer n, prove that
   $$1 + \frac{1}{\sqrt{2^3}} + \frac{1}{\sqrt{3^3}} + \ldots + \frac{1}{\sqrt{n^3}} < 3.$$

5. Use of Contrapositive Inference (反証法)
(Assume the truth of a statement which is opposite to what is required, and then prove that it leads to contradiction.)
(i.e., instead of proving P => Q, we try to prove that ~Q => ~P.)

Ex. 35: Given that a, b, and c are real numbers such that $abc > 0$, $a+b+c > 0$ and $ab+bc+ca > 0$, prove that a, b and c are all positive.

Ex. 36: Prove that if a and b are real numbers such that $a^3 + b^3 = 2$, then $a+b \le 2$.

Ex. 37: If a, b, c, A, B, C are real numbers satisfying $aC - 2bB + cA = 0$ and $ac - b^2 > 0$, prove that $AC - B^2 \le 0$.

6. Change of Variable (代換法)
(Introduction of a new variable.)

Ex. 38: Given that a and b are non-zero real numbers such that $a^2 + b^2 = 1$, Prove that
$$(a^2 + \frac{1}{a^2})(b^2 + \frac{1}{b^2}) \ge \frac{25}{4}.$$

Ex. 39: Given that $a^2 + b^2 \le 1$, prove that $-5 \le (3a^2 + 8ab - 3b^2) \le 5$.

Ex. 40 If a, b, c are positive real numbers, prove that
$$(b+c-a)(c+a-b)(a+b-c) \le abc.$$

Ex. 41: If a, b, c are positive real numbers, and $a + b + c \ge abc$, prove that among $\frac{2}{a}+\frac{3}{b}+\frac{6}{c} \ge 6$, $\frac{2}{b}+\frac{3}{c}+\frac{6}{a} \ge 6$ and $\frac{2}{c}+\frac{3}{a}+\frac{6}{b} \ge 6$, at least two of them are correct.

7. Mathematical Induction (數學歸納法)
(Use in proving inequalities involving a variable n, a positive integer.)

Ex. 42: Given that n is a positive integer, prove that
$$1 + \frac{1}{2} + \frac{1}{3} + \ldots + \frac{1}{2^n} > \frac{n}{2}.$$

Ex. 43: If $x_1, x_2, \ldots, x_n$ are all real numbers in the interval (0, 1) and their sum is equal to S_n, prove that for $n \ge 2$, $(1 + x_1)(1 + x_2)\ldots(1 + x_n) > 1 + S_n$.

Ex. 44: For any positive integer $n \ge 5$, prove that $2^n > n^2$.

Ex. 45: Suppose $x>0$ and [x] denotes the integral part of x, prove that
$$[x] + \frac{[2x]}{2} + \ldots + \frac{[nx]}{n} \le [nx]$$
for any positive integral value of n.